CEUE203 – Object Oriented Programming

Practical – 6 (Part A) | Student ID: 25IT037

1. Remote Control

Code

```
interface Switchable {
    void on();
 void off();
    default void toggle() { on(); }
}
class Fan implements Switchable {     public void
 on() { System.out.println("Fan ON"); }     public void
 off() { System.out.println("Fan OFF"); }
}
class Light implements Switchable {     public void
 on() { System.out.println("Light ON"); }     public void
 off() { System.out.println("Light OFF"); }
}
interface Permission {
    boolean maySwitchOn(Switchable d, int hour);
}
public class RemoteControl {
    public static void main(String[] args) {
        Switchable[] devices = {new Fan(), new Light()};
 for (Switchable d : devices) d.toggle();
 Permission a = new Permission() {
            public boolean maySwitchOn(Switchable d, int h) { return h >= 6 && h <= 22; }
        };
        Permission b = (d, h) -> h >= 6 && h <= 22;
        System.out.println("Anonymous: " + a.maySwitchOn(devices[0], 10));
        System.out.println("Lambda: " + b.maySwitchOn(devices[1], 23));
    } }
```

Output

Fan ON
Light ON
Anonymous: true
Lambda: false

2. Notification Senders

Code

```
@FunctionalInterface
interface Notifier { void send(String msg); }
 interface Urgent {} class Email
 implements Urgent {} class SMS {} public
 class NotificationDemo {     public static
 void main(String[] args) {
        Notifier email = msg -> System.out.println("Email: " + msg);
        Notifier sms = msg -> System.out.println("SMS: " + msg);
        Notifier[] senders = {email, sms};
 boolean[] urgent = {true, false};
 String msg = "Exam reminder";         for
 (int i = 0; i < senders.length; i++) {
 senders[i].send(msg);
            if (urgent[i]) senders[i].send(msg);
```

```
    }
} }
```

Output

Email: Exam reminder
Email: Exam reminder
SMS: Exam reminder

3. Discount Engine – Advanced Learners

Code

```
package discount; import
java.util.*;
interface Rule { double apply(double price); } public
class DiscountEngine {
   public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
      List<Double> prices = Arrays.asList(100.0, 250.0, 500.0);
      System.out.println("1. 10% discount");
System.out.println("2. 20% discount");
      int ch = sc.nextInt();
      Rule rule = ch == 1 ? p -> p * 0.90 : p -> p * 0.80;
for (double price : prices)
         System.out.println(price + " -> " + rule.apply(price));
  } }
```

Output

1. 10% discount
2. 20% discount 1
100.0 -> 90.0
250.0 -> 225.0
500.0 -> 450.0

Learning Outcome

I learned interfaces, default methods, functional interfaces, lambdas, anonymous classes, marker interfaces, and lambda-based rule processing in Java.